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# BALLARD POWER SYSTEMS

MGT 663 Strategic Opportunities in Impact Enterprise, Fall 2020

**Sustainable Transportation**

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# Introduction and Company Overview

As climate change increasingly promises impending doom, Ballard Power Systems Inc. (BLDP) exists to satisfy the societal need of reducing transportation emissions. The transport sector as a whole is responsible for about 8 billion metric tons of $CO_2e$ annually, and there is growing pressure to reverse this (Ritchie, et. al, 2020). BLDP's hydrogen fuel cells can power zero emission vehicles with only water as a byproduct, completely eliminating transportation's toxic fumes that come from standard internal combustion engine vehicles.

BLDP is headquartered in Burnaby, British Columbia and is a developer and manufacturer of hydrogen fuel cell stack hardware for buses, trains, personal vehicles, and the heavy transport sector. They also partner with HDF Energy to produce large, stationary fuel cells to power plants. BLDP went public on the Toronto Stock Exchange in 1993 and was listed on the NASDAQ in 1995. They currently employ approximately 650 people, have sold more than 670 MW worth of electrical power generation capacity in fuel cells, and have a market capitalization of $3.5 billion. BLDP is majority owned by Daimler AG, Ford Motor Co, and Weichai Power. In addition to manufactured goods, BLDP also offers an array of technical services ranging from component design to tech licensing and tech transfer. BLDP is the most highly ranked public company in the fuel cell systems space from a Corporate Social Responsibility lens (CSR Hub).

BLDP develops and manufactures Proton Exchange Membrane Fuel Cell (PEMFC) technology that uses hydrogen as a fuel by combining it with oxygen from the air, with water as a byproduct, to produce electricity (through the liberation of electrons when $H_2$ is split upon contact with the PEM). PEMFCs are constructed using a fluoropolymer (think Teflon) backbone making them mechanically and chemically robust. PEMFCs can operate at low temperatures (-80 °F), are physically lightweight, ramp up to full power rapidly, and can achieve thermodynamic efficiencies greater than internal combustion engines. Hydrogen fuel cells turn out to be more efficient than electric vehicles for heavy duty applications such as buses and semi-trucks.

The hydrogen market is growing rapidly, and "is currently enjoying unprecedented political and business momentum" (IEA, 2019). Given the current state of the market, BLDP is well positioned for continued growth. Their buses are already used for public transport across many countries in Europe, and they are breaking into China's mass transit. In this report, the focus will be on the bus transportation applications of Ballard's technology.

# Industry Analysis

## Sector Definition

This analysis focuses on fuel cells manufacturers producing fuel cells for non-passenger vehicle transportation sector applications. Here, *transportation sector* refers to electric (motor-powered) vehicles (and does not include the stationary facilities where they are produced, maintained, and stationed) which source their electricity predominantly from fuel cells (rather than batteries or a hybrid technology). Because this sector technically includes a diversity of fuel cell technologies, vehicles in this sector can accept several different primary energy (fuel) sources. Those include compressed hydrogen gas, compressed and liquefied natural gas, compressed and liquefied petroleum gas, compressed propane, methanol, ethanol, and ammonia (all of these aside from hydrogen and ammonia will generate CO2 as a reaction product).

Ballard offers PEMFC technology, which is compatible with hydrogen only. Although there is an entire value chain associated with the fuel cell itself, this analysis specifically investigates "plug-and-play" fuel cell stack units encapsulated in a single enclosure which gets installed into the vehicle. These companies sell directly to vehicle (auto, bus, train, forklift, etc.) manufacturers, and are thus strictly B2B. Ballard currently markets its PEMFC for use in transit buses, passenger vehicle applications, commuter rail, truck and tractor trailers, material handling (fork lifts), UAV's, "Critical Infrastructure" (back-up power systems), and Marine power markets (Ballard-Markets, 2020).

## Current Market Structure


***Figure 1. Companies in the Transportation Fuel Cell Space***

## TAM SAM SOM Analysis

- Here we will define the T.A.M. (Total Available Market) as the entire alternative energy and alternative fuels transportation market. This is valued at $593B/year (by 2022), according to Zion Market Research (ZION, 2018)
- We will define the S.A.M. (Serviceable Available Market) as including the following sectors F.C. Busses, F.C. Trains, F.C. Ships, F.C. Forklifts, F.C. box trucks and tractor trailers, FC UAV (because these are vehicles types which Ballard actively supports with their PEMFC unit. This market might be valued at $1.1B/year (10% of total global fuel cell market), according to Grand View Research (Grand View Research, 2020)
- Ballard's S.O.M. (Serviceable Obtainable Market) - Assumed average 50% Mkt share of the PEMFC market for each of these transport markets – $300-500M/year in 2019. Because PEMFC (and F.C.'s in general) are expected to grow exponentially over the next decade, these figures are expected to grow significantly.

## Direct Competitors

Although all of the large PEMFC manufacturers have some extent of differentiation, there are still a handful of manufacturers that would be considered direct competitors. These include groups such as: Plug Power, PowerCell Sweden, A.F.C. Power, and Hydrogenics.

Some examples of these differentiating factors include tailoring to niche vehicle markets (Ballard makes fuel cells for forklifts), specific subsets of large vehicle market (Hydrogenics tailors to light rail, Ballard tailors to passenger rail) (Hydrogenics, 2019). In the marine sector, PowerCell advertises ferries and small vessels, Ballard also mentions ferries, but also barges and cruise ships (Ballard, 2020).

Although it might be difficult for Ballard expand market share, there is great potential lying in growth of the industry itself. The Road Map to a US Hydrogen Economy, between today (2020) and 2030, predicts a 10X increase in hydrogen-powered material handling, and a 500X increase in hydrogen passenger vehicle and truck sales (FC and H2 Energy Association, 2020). If this forecast has any merit, Ballard (and their competitors) have a bright future.

## Sector history, growth, and current status.

Although battery-electric and fuel cell-electric vehicles may seem like a modern advent, they have origins dating back to the beginnings of the modern vehicle. The first "crude" electric vehicle was developed in 1832, and the first fuel cell was developed by an experimentalist, Robert Grove, in 1838 (Schoenbein, 1839; U.S. DOE, 2020). Grove's "gas battery" was a platinum electrode submerged in sulfuric acid electrolyte fuel cell, which converted hydrogen and oxygen into electrical current, water, and heat. It was not until 1893 was a sound theoretical understanding of the fuel cell was established, by Friedrich Wilhelm Ostwald (Smithsonian Institute-1, 2004). It was not until the 1880's and 1890's did reliable fuel cell designs emerge. They gained popularity in use with Mond-gas (gasified coal, commonly called *syngas* today, primarily composed of CO2 and H2), named after the chemist and industrialist Ludwig Mond (Smithsonian Institute-2, 2004). In the 1930's Francis Bacon developed the alkali electrolyte fuel cell (distinct from the previously developed acid fuel cells), similar technology of which was used in the Apollo spacecraft fuel cells (Smithsonian Insitute-2, 2004). The first Proton Exchange Membrane (P.E.M.) fuel cells were developed by Thomas Grubb and Leornard Neidrach. These lightweight, more reliable fuel cells replaced alkali fuel cells in subsequent NASA missions. But it wasn't until the 1980's did P.E.M. fuel cells become practical for modern stationary and motive power generation applications.

Ballard Power Systems was one of the first companies to commercialize the P.E.M., in 1989 (Smithsonian Institute-2, 2004). In 2008, Ballard split part of the company which focused on personal vehicle fuel cells, the company was called Automotive Fuel Cell Corporation (AFCC) (Siemens Digital Industries, 2017). In 2018, after having a clearer vision of industry trends (in mobile fuel cell power applications), Ballard required AFCC (Ballard, 2020). Ballard's I.P.O. on the Toronto Stock Exchange was in 1993 and was subsequently listed on the NASDAQ in 1995.

In their nascency, the bus fuel cell market grew in fits and starts through federal research funding opportunities, regional pilot programs and industry pilot programs. In the mid 1990’s, Ballard went outside their PEM comfort zone to work on developing a methanol fuel cell bus for a DOE grant (Industry Canada, 2003). Around the same time, they put PEMFC’s into a handful of busses for the Chicago and Vancouver transit authorities. Not to long after, UTC Power did the same with Georgetown University buses. UTC appeared to have the upper hand for a while after establishing contracts with Thor bus lines and VanHool bus lines – which it expanded through the mid 2000’s. Ballard sold 30 units to Mercedes-Benz bus division for installation in their Citaro model as a part of the European Fuel Cell Bus Project – establishing a name for Ballard in the EU (Industry Canada, 2003). In 2008, UTC again outsold Ballard, in the largest sale of fuel cells for buses (120 units) to Oakland, CA’s AC Transit (UTC Power, 2012). Initiatives and pilot programs expanded rapidly in the 2010’s (with international efforts, such as the Hydrogen Bus Alliance), allowing Ballard to expand market share, establishing contracts with manufacturers such as NewFlyer, El Dorado, Solaris, and VanHool (Hydrogen and Fuel Cells Canada, 2008).

### Overview of key technologies or technological developments

There are three common types of batteries which have been used in vehicle applications: lead-acid batteries, nickel-metal hydride batteries, and lithium-ion batteries. Lithium-ion batteries have been the focus of most recent battery electric vehicles due to their significant power-to-weight advantage over the other types. Even with modern technology, battery electric vehicles are less economical (on a cost and weight basis) than a fuel cell electric vehicle (U.S. DOE EERE, 2020).

Although there are more than 20 types of fuel cells, only three types currently have commercial relevance for vehicle applications; PEMFC's, Solid Oxide Fuel Cells (SOFC's), and Molten-Carbonate Fuel Cells (MCFC's). Of relevance to this discussion, these three types vary in the fuels which they can handle. PEMFC's can only use pure hydrogen, while SOFC's and MCFC's operate at high enough temperatures to reform a variety of hydrogen carriers and hydrocarbons to generate the hydrogen gas required to generate electrical current at the anode. PEMFC's are currently the most developed technology for vehicle applications.

## Description of the industry value chain

The transportation sector is renowned for having one of the most complex multi-tiered supply chains. With continual reduction in the barriers to market and communicate, and with continual increases in technology and sophistication, it is unlikely that the supply chain will simplify. In fact, the opposite will likely happen – vendors and suppliers with be able to further specialize. When it comes to large publicly traded companies – the M&A game muddles this picture, but it is often cheaper and risk-adverse to outsource part manufacture. Other trends noted by Deloitte are the transitioning to Manufacturing 4.0 (a wholistic adoption of smart planning, control, and smart strategy), the transition towards incorporating autonomous vehicle attributes, and most notably batteries and electrification as an emerging market (Deloitte, 2017).

Ballard provides value to its suppliers through meaningful relationships and the potential for exponential growth, Ballard generally provides value to its customers by enabling decarbonization of a complex energy-intense sector. Ballard's role as a co-developer with its customers might be the attribute which makes it so valuable. This co-creation process is resource (man-hour) intensive and requires a company of a size like Ballard to successfully achieve (Ballard, 2020).

## Sector Structure

The fuel cell electric transportation market (from the focal point of the fuel cell stack itself) is primarily a B2B market (there are few companies manufacturing fuel cell stacks designed for transportation applications sold directly at end user markets). The industry is far from mature, but fuel cells, particularly the type of fuel cells which Ballard manufactures (proton exchange membrane), have been commercially available in stationary power applications since the mid 1980s, and in lightweight passenger vehicle applications since the early 2000s (Smithsonian Institution, 2004; Andu ́jar & Segura, 2009). This state of development is relevant because it important for establishing context for the threat of competition. When looking at the industry, although there are a few dozen active companies in the space, there are only a handful which have been around for long enough, have grown large enough, and have developed a brand recognition that gives them access to establishing contracts with large transportation vehicle manufacturers and thus garner market share. The buyers here are somewhat price-sensitive, because no-longer are the days they can get away with only offering a $60k fuel cell vehicle inaccessible to the masses. There is some leverage held by the fuel cell stack manufacturer with the best reputation; they might be able to get away with decently high prices for bus fuel cell stacks. Design of modern vehicles is complex because the F.C. manufacturer must work directly with the vehicle manufacturer to tailor design of the stack enclosure and enclosure mounting system; there will be very high switching costs. Because of the handful of large PEMFC manufacturers which have established brand recognition, there are high barriers to entry for any other players. It would likely take Tesla-scale investment to come out of left field and become an active threat to the main PEMFC players.

## Five Forces Analysis

### Rivalry Among Existing (Current Competitors)

There are a few threats from current competitors that are other large companies who manufacture PEMFCs for mobile applications. These include Plug Power, A.F.C. Energy, and PowerCell Sweden, for example. These companies could in theory out-develop Ballard Power to produce a more efficient, lower cost, more compact fuel cell stack.

Much of Ballard's success depends on the way vehicle power sources trend – whether battery electrification outcompetes fuel cell electrification. Although there is potential for these trends to shift, from a physics perspective, it is unlikely that battery electric will be able to outcompete fuel cells in heavy transport and public transit (on a weight and volume basis).

Another trend that in theory could affect the fuel cell transport industry is the production of low carbon and carbon-negative biofuels. From the scientific community, it is generally understood that biofuels might ideally be used predominantly as a transition fuel. Biofuels have great potential for decarbonizing jet air travel – where widespread electrification is not generally feasible for many decades. From an engineering perspective, many biofuels work as drop-in fuels and thus allow for a more frictionless decrease in the carbon intensity of existing infrastructure. Even though that is the case, there will still be a tremendous effort to begin transitioning our infrastructure – to which fuel cells offer a good long-term sustainable solution.

## Potential (Threat of) New Entrants

As mentioned above, for another company to be considered a new entrant direct threat, they would need to provide a lower price product, a high efficiency/longer range product, or a product that was lighter weight, not to mention overcoming great cost and brand recognition barriers.

Likely the greatest threat to Ballard is one of the deep-pocketed companies tangentially in the space (such as Tesla), who could pivot or form a division which will manufacture the same PEMFC.

Another alternative is the wide-spread shift or adoption of one of the fuel cell technologies which accepts fuels other than hydrogen gas. These other technologies have the advantage of fuel flexibility to be able to handle other gases like natural gas, propane, or liquid methanol. The challenge with these other fuels is they will release $CO_2$ as a byproduct and are thus not necessarily a long-term ideal carbon-zero fuel.

## (Bargaining) Power of the Supplier

Because the fuel cell industry could be considered to be in its infancy, there are currently a finite number of manufacturers of the components used in Ballard's fuel cells. Thus, Ballard has little leverage over them. That said, building and maintaining close relationships with Ballard will be very valuable for vendors as this industry grows exponentially.

## (Bargaining) Power of the Buyer

Because there are currently only a few manufacturers of this technology, the buyers only have some power. Particularly because Ballard appears to be the furthest along in developing a wide variety of vehicle-tailored stack enclosure and mounting systems, which required a tremendous investment on the part of both Ballard and the customer; thus, there are very high switching fees. Because Ballard is trying to exponentially grow its business, it does need to treat the customer with good service.

### Threat of Substitute Products

Substitute products were touched on briefly above. Biofuels pose a near-term threat. Truck and bus manufacturers need to weigh the pros and cons of different power sources for their vehicles. There are high switching costs, so generally a path chosen will be stuck to for a number of years. In this case, it might mean investing in modified engines which can be fed renewable natural gases or renewable D.M.E. (generally modifications are not needed for renewable diesel). Although from a materials science perspective it will be very technically difficult for battery densification and light-weighting technology to become more attractive than fuel cells for heavy transport applications, in light-duty applications this (BEVs out trending FCEVs) is a major concern.

## Recent and Likely Future Changes

Beyond consumer trends, policy incentives could be a huge factor influencing widespread adoption. In general, however, most carbon pricing, taxing, or crediting schemes strongly favor the development of fuel cells.

The past decade has seen a windfall for electrification over biofuels and transition from gasoline and diesel fueled vehicles to electric could be considered a megatrend. Electrified vehicles will likely see continued exponential growth for the coming decades.

Due to weight and volume factors limiting the range of battery electric vehicles, fuel cells are currently viewed as *the* go-to electrification technology.

If nothing else demonstrates this trend, the 30X increase in Ballard Power's stock price is a testament to the growth curve being experienced in this industry.

## Key success factors

- Brand recognition / marketing strategy
- Commercial experience (our system has been installed and running w/o maintenance in 10,000 vehicles for 5 years)
- Financial viability/cost savings story for manufacturers
- Large enough that company will be around to service & support product decades into the future
- Product compatibility with your vehicle

# Strategic Analysis

Ballard primarily works in the design, development, manufacture, sale, and service of Proton Exchange Membrane (P.E.M.) fuel cell products for various markets like transit bus, rail, automotive, truck, rail, UAV, material handling, critical infrastructure, and marine. Ballard has a patented P.E.M. fuel cell technology. It operates to build shareholder value by developing, manufacturing, selling, and servicing zero-emission P.E.M. fuel cell technology-based products.

BLDP's business strategy is to build shareholder value through the sale and service of power products and deliver technology solutions[1].

1. Within Power Products, the focus is on meeting customers' power needs by delivering high value, high reliability, high quality, and clean energy power products that reduce customer costs and risks[1].
2. Within Technology Solutions, the focus is on enabling customers to solve their technical and business challenges and accelerate their fuel cell programs by delivering customized, high value, bundled technology solutions[1].

Ballard has increased its efforts to grow the business in China since 2015. The factors like continued urbanization, infrastructure development, air pollution challenges, increased adoption of electric vehicles, etc., have provided Ballard's a unique opportunity to tap this growing market by providing zero and low-emission motive solutions.

## Mission, Purpose, and Values

***Vision:*** To deliver fuel cell power for a sustainable planet[1].

***Mission:*** Use BLDP's fuel cell expertise to deliver valuable and innovative solutions to the customers globally, create rewarding opportunities for BLDP, provide extraordinary value to shareholders, and power the hydrogen society[1].

***Values:*** Ballard Power defines how it carries out business based on five core values. In addition to safety and innovation, the five key cultural values of BLDP are[2]:

1. *Listen and Deliver* – Listen to customers, understand their business, and deliver innovative and valuable solutions for lasting partnerships.
2. *Quality Always* – Deliver quality in everything BLDP does.
3. *Inspire Excellence* – Live with integrity, passion, urgency, agility, and humility.
4. *Row Together* – Achieve success through respect, trust, and collaboration; and
5. *Own It* – Step up, take ownership of results, and trust others to do the same.

## Products, technologies, or services

Ballard offers four distinct fuel cell-powered products for applications in heavy-duty vehicles, marine vehicles, Unmanned aerial vehicles (UAV), and power back up. The four product categories are listed below:

1. **Fuel cell stacks**: Fuel cell stacks are the primary component of a fuel cell system. O.E.M.s and system integrators use these stacks to produce fuel cell systems for power solutions. Ballard offers both air-cooled and liquid cooled P.E.M. fuel cell stacks mobility and stationary applications
2. **Fuel cell modules:** Ballard designs and builds fuel cell modules using their fuel stack and procuring the balance of plant components. These modules are then provided to system integrators and O.E.M.s. These modules have 30 kW to 100 kW net power and are used in heavy-duty vehicles like transit buses, trucks, and light rails.
3. **Marine System**: Ballard offers 200 kW fuel cell modules that are used provide zero-emission propulsion power to marine vessels
4. **Backup power systems:** Ballard offers a complete stationary fuel cell system that provides backup power to critical infrastructure.

Ballard also offers customer services like training, commissioning, engineering services, application engineering, and after-sales service packages.

## Position in the value chain

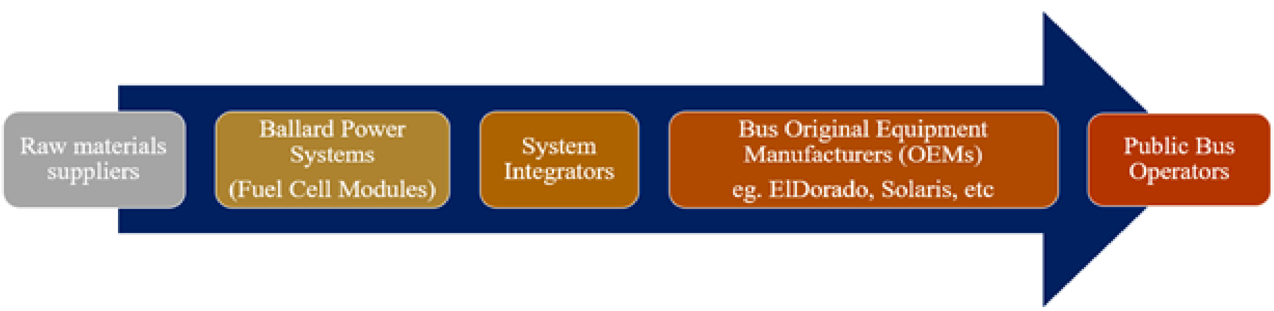



*Figure 2: Position of Ballard in the entire value chain*

Ballard power systems is positioned towards the tail-end of the entire value chain of electric buses from extraction to usage. A particular company manufactures any electric bus operated on the road. This manufacturing company is called the original equipment manufacturer (O.E.M.). Each electric bus is made up of many parts. Some parts like bus chassis, body, etc. are manufactured by O.E.M.'s themselves, and some parts like engine, battery are sourced from other companies. Fuel cell module is one such part that is sourced from other companies.

Fuel cell suppliers fall into two groups: (1) those that manufacture and integrate the entire fuel cell system and (2) those that manufacture only fuel cells to be used by others to integrate the whole system. BLDP supplies both fuel cells as individual components to integrators and the entire fuel cell system directly to O.E.M.'s. The value chain starts with the supply of raw materials for the manufacturing of fuel cells by BLDP.

## Primary target market(s)

BLDP designs and produces fuel cell products in eight different market segments: transit bus, rail, automotive, truck, UAV, material handling, critical infrastructure, and marine.

1. **Transit Bus:** Zero-emission buses are a critical factor for the transition towards a cleaner public transportation sector. Hydrogen-powered fuel cell electric bus is a perfect replacement for diesel-powered vehicles with complete route flexibility, short refueling time, and similar depot space utilization. Ballard has supplied fuel cell technology to more than 1000 electric buses globally that have operated more than 50 million kilometers to date.
2. **Rail:** Ballard designs and manufactures fuel cell modules to supply train and tram manufacturers. Ballard motive modules power the world's first hydrogen fuel cell-powered fixed rail electric tram in China (Ballard 2020-Rail).
3. **Automotive:** Ballard collaborates with leading automotive O.E.M.'s, including Volkswagen AG, Audi, Daimler AG, Ford Motor Company, and Honda Motor Company. Ballard supplies fuel cell stacks and tailors its product development to the specific requirements of customers. Ballard has its range of proprietary anode and cathode catalysts, electrodes, sublayers, membrane additives, seals, and plate materials from which they quickly design and build new products (Ballard 2020-Automotive).
4. **Trucks:** Fuel cell electric trucks can be used for different applications of long-range, heavy payloads like drayage trucks, refuse trucks, cement trucks, delivery trucks, utility trucks, and mining trucks. Ballard has supplied the fuel cell modules from 30 to 200+ kilowatts for medium and heavy-duty trucks (more than 2,200 trucks) deployed in China, Europe, and North America to provide power (Ballard 2020-Truck).
5. **UAV:** Ballard sold its Unmanned Aerial Vehicle business to Honeywell on October 15, 2020.
6. **Material Handling:** Within the material handling industry, Ballard provides fuel cell stacks to power forklifts. Ballard supplies fuel cell stacks to systems integrators and material handling equipment manufacturers for integration into forklifts. Ballard also gives these stacks to Plug Power Inc., which holds the largest share of the fuel cell material handling equipment market (Ballard 2020-Material Handling).
7. **Critical infrastructure:** Ballard offers a power backup solution for power outages at critical infrastructure sites, including radio towers, rail, data centers, telecommunication equipment, etc. Fuel cell-powered backup systems are a reliable and cost-effective alternative to battery and diesel generators. It has deployed this technology for more than ten years in thousands of systems worldwide, with millions of operating hours (Ballard 2020-Critical Infrastructure).
8. **Marine:** Ballard provides modular, scalable fuel cell systems for ferries, barges, and cruise ships. A fuel cell vessel uses a hybrid electric system (both fuel cells and batteries) to provide efficient zero-emission power. Usage of fuel cell technology in the marine industry is a relatively new concept, and the market is yet to grow. Ballard is also involved in developing the world's first sea-going renewables-powered car and passenger ferry (Ballard 2020-Marine).

## The value proposition to the customer.

The fuel cell electric buses (FCEB) have similar benefits as that of battery buses with extended advantages like long driving range, reduced vehicle weight, route flexibility (no roadside infrastructure), and short refueling times. FCEB is a zero-emission solution that emits nothing but water vapor at the tailpipe of the buses. FCEB can further reduce CO2 emissions by producing hydrogen from a low carbon source. Additionally, FCEB offers enhanced comfort, reduced noise and vibrations levels, and a low carbon transportation system solution.

All these advantages of an FCEB make up a value proposition for any O.E.M. to source fuel cell systems, thereby creating an opportunity for BLDP to be the supplier. BLDP claims superior fuel economy, range, and similar operating costs as diesel and C.N.G. powered buses, as the value proposition of fuel cell buses.

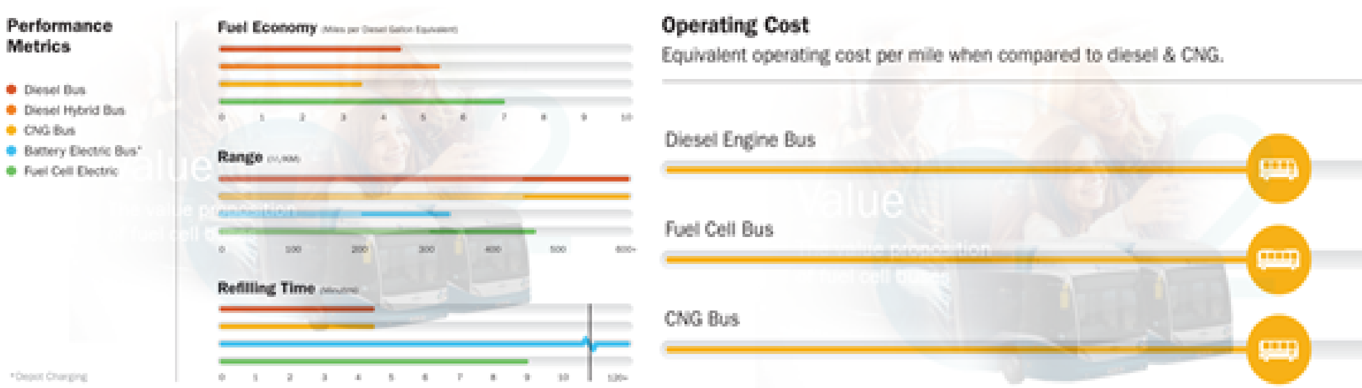



*Figure 3: Value Proposition (Ballard, 2020).*

## Competitor Analysis

There are five types of fuel cell technologies available at present. Each type of technology has different operating temperatures, stack sizes, electrical efficiency, and applications, along with their advantages and challenges. Different companies working in the fuel cell industry use different technologies. Ballard's four significant competitors in the fuel cell industry include Plug Power, A.F.C. Energy, Bloom Energy Corporation, and Hydrogenics.

*Table 1: Competitor Analysis*

| | Ballard | Plug Power | A.F.C. Energy | Bloom Energy | Hydrogenics |
|---|---|---|---|---|---|
| **Founding Date** | 1979 | 1997 | 2006 | 2001 | 1995 |
| **Headquarter** | Burnaby, Canada | Latham, US | Cranleigh, Great Britain | San Jose, US | Mississauga, Canada |

| | | | | | |
|---|---|---|---|---|---|
| **Type** | Public | Public | Public | Public | A subsidiary of Cummins Inc. |
| **Work** | Design, development, manufacture, sale, and service of P.E.M. fuel cell products in various markets. | Design, develop and sell hydrogen-based fuel-cell products | Develops low-cost alkaline fuel cell technology | Manufacture and sale of solid-oxide fuel cell systems for on-site power generation | Design, manufacture, build and install industrial and commercial Hydrogen generation, fuel cells solutions |
| **Application** | Transit bus, rail, automotive, truck, and marine applications, as well as UAV, material handling, and backup power for critical infrastructure | Fleet vehicles (trucks), ground support equipment, UAV's industrial robotics, material handling, and backup power | Focuses on large-scale industrial applications to provide clean electricity on-demand. It also works in the mobility, construction sector. Withing mobility, their focus is on Electric vehicle cars | Stationary power generation platform | Electric vehicles such as transit buses, commercial fleets, utility vehicles and electric lift trucks. It also works in fueling stations, electrical power plants, U.P.S. systems, and other sectors. |
| **Employees** | 900 | 835 | 40 | 1524 | 140 |
| **Enterprise value ($) (At close: 23 October Source: Yahoo Finance)** | 3.76 Billion | 5.77 Billion | 148.82 Million | 3.22 Billion | NA (applicable only for public companies) |
| **Market Capitalization ($) (At close: 23 October Source: Yahoo Finance)** | 4.16 Billion | 6.04 Billion | 151.6 Million | 2.25 Billion | NA |
| **Patents (Foreign) (Source: Craft.co 2020)** | 100 (Mar, 2020) | NA | N/A | 124 (FY, 2019) | N/A |
| **Patents (US) (Source: Craft.co 2020)** | 43 (Mar, 2020) | 109 (FY, 2019) | N/A | 240 (FY, 2019) | N/A |

Although many companies are now working in the hydrogen fuel cell industry, very few of them work extensively in the transportation sector. Amongst all the competitors, PlugPower is the closest and major competitor for Ballard as PlugPower operates in the various markets in the transportation sector and has an established network.

## Source of competitive advantage and sustainability thereof

With its 40 years of rich industry experience in manufacturing fuel cell products, Ballard is a well-positioned company with unique competitive advantages. These advantages include top-tier long-term customers, strategic shareholders, and leading cutting-edge technology (Ballard 2020-Competitive Advantage).

**Technology**

Ballard is a technology leader in clean energy fuel cell innovation and engineering with decades of industry presence. It has invested over $1 billion in fuel cell product development since its inception. It has a strong innovation policy portfolio consisting of ~200 patents/applications with access to ~1,400 patents/applications while designing optimized membrane electrode assemblies (M.E.A.) and fuel cells (Ballard 2020-Competitive Advantage).

**Collaboration:**

Over the past many years, Ballard has created extensive collaborations with Tier 1 automotive manufacturers to enhance its product durability, performance, and reduced product cost simultaneously. Some features of Ballard's fuel cell products are high fuel efficiency, low noise and vibration, compact size, quick response to electrical demand changes, and modular design (Ballard 2020-Competitive Advantage). Ballard's collaboration in various market segments are listed below:

*Table 2: Major collaborations*

| Transit Bus | Rail | Truck | Automotive | Marine | Material Handling |
|---|---|---|---|---|---|
| **ElDorado –** manufactures light- and medium-duty mid-size commercial buses for the North American market | **CRRC -** world's largest supplier of rail transit equipment | **MAHLE -** leading supplier to the commercial vehicle and automotive industry | **AUDI AG -** for the design and manufacture of world-leading, next-generation fuel cell stacks | **ABB –** for development of megawatt (M.W.) scale proton exchange membrane (P.E.M.) fuel cell power systems for the marine market | **Plug Power** – to deploy Ballard fuel cell stacks |
| **New Flyer –** largest heavy-duty transit bus manufacturer in North America | | | | | |
| **Solaris –** major European producer of city, intercity and special-purpose buses as well as low-floor trams | **Siemens -** largest industrial manufacturing | | | | |
| **VanHool –** 4th largest bus manufacturer in Europe | | | | | |

**Manufacturing:**

Ballard operates three facilities in Burnaby, Canada, with a combined footprint area of over 240,000 ft² and capacity to manufacture over 1 million M.E.A.'s and 10,000 stacks annually. Ballard's automated and continuous manufacturing process enables it to produce M.E.A.'s, stack assembly, module integration, and complete system testing (Ballard 2020-Competitive Advantage).

**Sustainability:**

Ballard in their 2019 E.S.G. Annual report, mentioned that they want to improve the company's performance in the following United Nations' Sustainable Development Goals:


- S.D.G. 7: Affordable and clean energy
- S.D.G. 9: Industry, innovation, and infrastructure
- S.D.G. 12: Responsible consumption and production
- S.D.G. 13: Climate action

Ballard is committed to reducing its environmental impact by 2020 and through the following actions:

1. **Minimize waste in manufacturing** by ethically sourcing materials, using materials made without any hazardous, banned, or toxic substances, manufacturing sustainably, reducing packaging as minimally as possible, using environmentally friendly materials, and producing long-service life products.
2. **Carbon Neutral by 2030** by continuously measuring, tracking, and monitoring the carbon footprint under various scopes (Scope 1,2 and 3). Ballard is also focusing on Life-Cycle Assessments of fuel cell stacks and modules to measure their carbon footprint "from cradle to gate." Total greenhouse gas emissions per - employee are down 16% from 2017 to 2018 (Ballard-ESG Report, 2019).
3. **Continually Reducing Energy Consumption**: Ballard has developed a target list of 42 initiatives to reduce energy use while working with BC Hydro (electrical utility) and others. As a result, electricity consumption decreased by 2.4 MWh during 2017-2018, and in 2019, the energy consumption per employee reduced by 19%, compared to 2018 (Ballard-ESG Report, 2019).
4. **Fuel Cell Refurbishing and Recycling:** Ballard recycles and refurbishes thousands of fuel cell stacks every after the fuel cells reach end-of-life to keep them out of the landfill

## Core competencies (tangible and intangible resources) leading to competitive advantage

Ballard is a leading manufacturer of hydrogen-based fuel cells for the transportation industry. Ballard's core competencies (tangible and intangible resources) that provide it a competitive advantage over other companies are listed below:

*Table 3: Core competencies of Ballard*

| | | |
|---|---|---|
| **Years of Experience** | 40 years | Ballard is one of the oldest companies to have been working in the fuel cell industry. They have rich experience in the entire process, from design to manufacturing to selling. |
| **Employees** | 900 | A blend of highly qualified employees drives the intellectual capital at Ballard. It has 2,300 person-years of experience with 211 Engineers, Scientists & Technologists in the company. The average tenure of their core engineering staff is 11 years.<br>**Educational Qualifications:** 30 PhD's, 48 Masters, 100 Bachelors & 33 Diplomas |
| **Patents & Applications** | 1400 | Ballard owns, licenses, and has access to ~1,400 patents & patent applications that give Ballard an advantage in terms of intellectual capital and property it owns. |
| **Stock exchange** | NASDAQ & TSX | Ballard has been a publicly listed company for over 25 years in NASDAQ and 27 years in TSX (Toronto stock exchange) |
| **Strategic Shareholders** | 4 Strategic shareholders | Ballard has four strategic shareholders: Weichai Power and Zhongshan Broad-Ocean Motor from China, along with Anglo American Platinum from South Africa and Nisshinbo Group from Japan.<br>The partnership with Weichai Power and Zhongshan Broad-Ocean Motor is intended to enhance Ballard's China strategy and establish itself at the forefront of China's low carbon transportation revolution. |
| **Application** | >5.5 Million M.E.A.'s produced and 670 MW fuel cell products delivered. | Ballard has supplied fuel cell products (modules, stacks, etc.) to various target markets. There are over 1000 transit buses, 2200 trucks, and 12000 forklifts in operation that use Ballard's fuel cell system.<br>Ballard is involved in four train projects in the rail markets and five ships in the marine market. |
| **Facility** | 50+ testing stations | Ballard has comprehensive testing and prototyping facilities, which can be scalable to produce a high volume of finished products. |

# SWOT Analysis

*Table 4: SWOT Analysis*

| | | |
|---|---|---|
| 1. Industry experience<br>2. Big network within the industry<br>3. Intellectual property (Patents and highly skilled employees)<br>4. Zero-emissions technology<br>5. Existing production infrastructure and R&D facilities | | 1. Inadequate investment in hydrogen infrastructure and limited hydrogen fuel stations<br>2. Still a developing sector<br>3. Operating at a loss historically<br>4. High hydrogen fuel costs |

| | | |
|---|---|---|
| 1. The growing fuel cell market<br>2. Opportunities in China and India<br>3. Collaboration with big O.E.M. companies and integrators<br>4. Not many competitors in the transportation sector<br>5. Growing environmental concerns among the general public<br>6. Improve the overall efficiency of the system |  | 1. Customer or production delays<br>2. Opportunities in China and India<br>3. Adverse macro-economic conditions for the International market, especially China<br>4. Changes in government subsidy and incentive programs for the China market<br>5. PlugPower overtaking market share the forklift market |

## Blue Ocean Strategy

Blue ocean strategy aims at developing a market segment and capture the new demand by making the market competition irrelevant. Adopting this blue ocean strategy will make Ballard a long-time sustainable brand/company. The actions that Ballard can take to create a value curve are shown below:

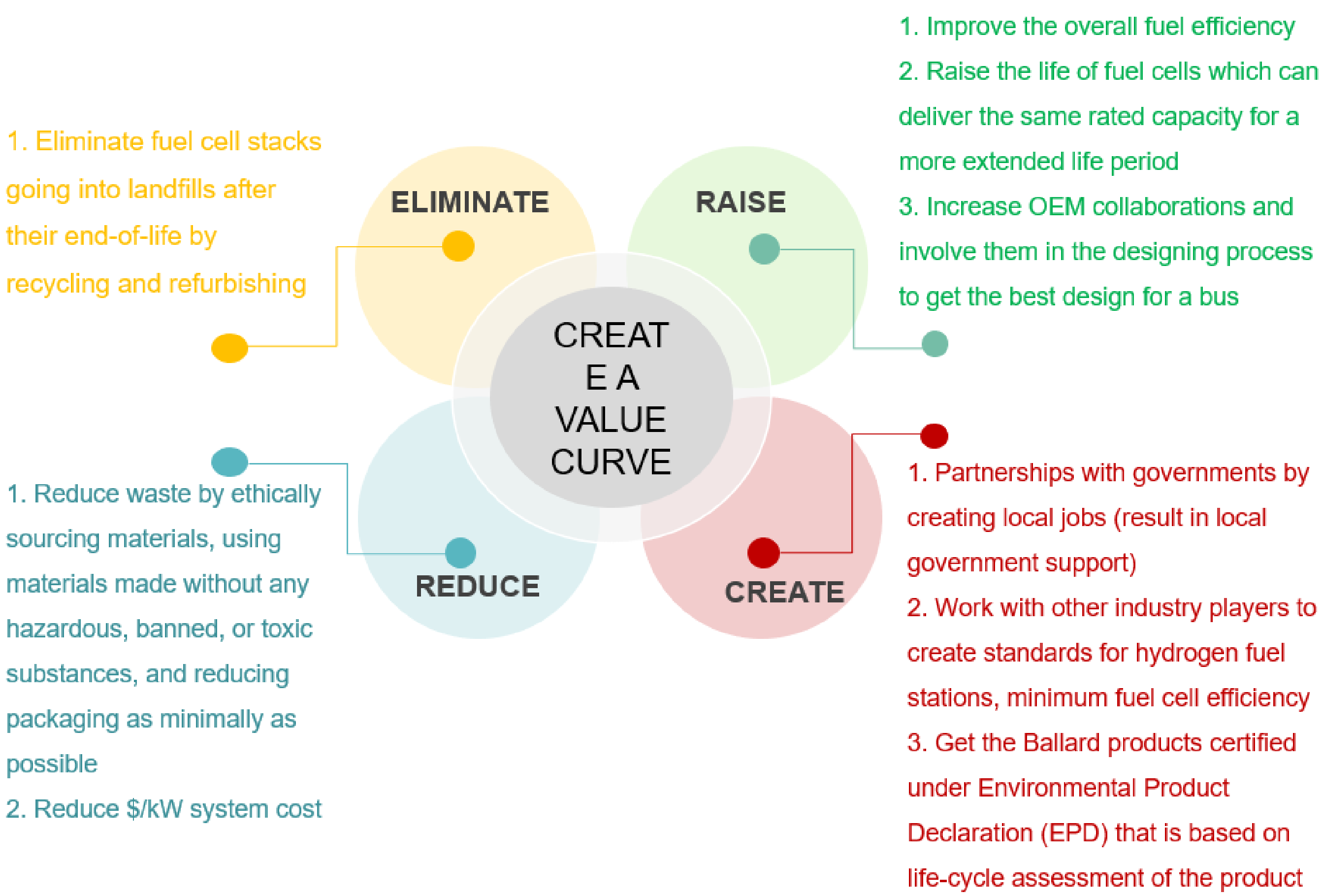


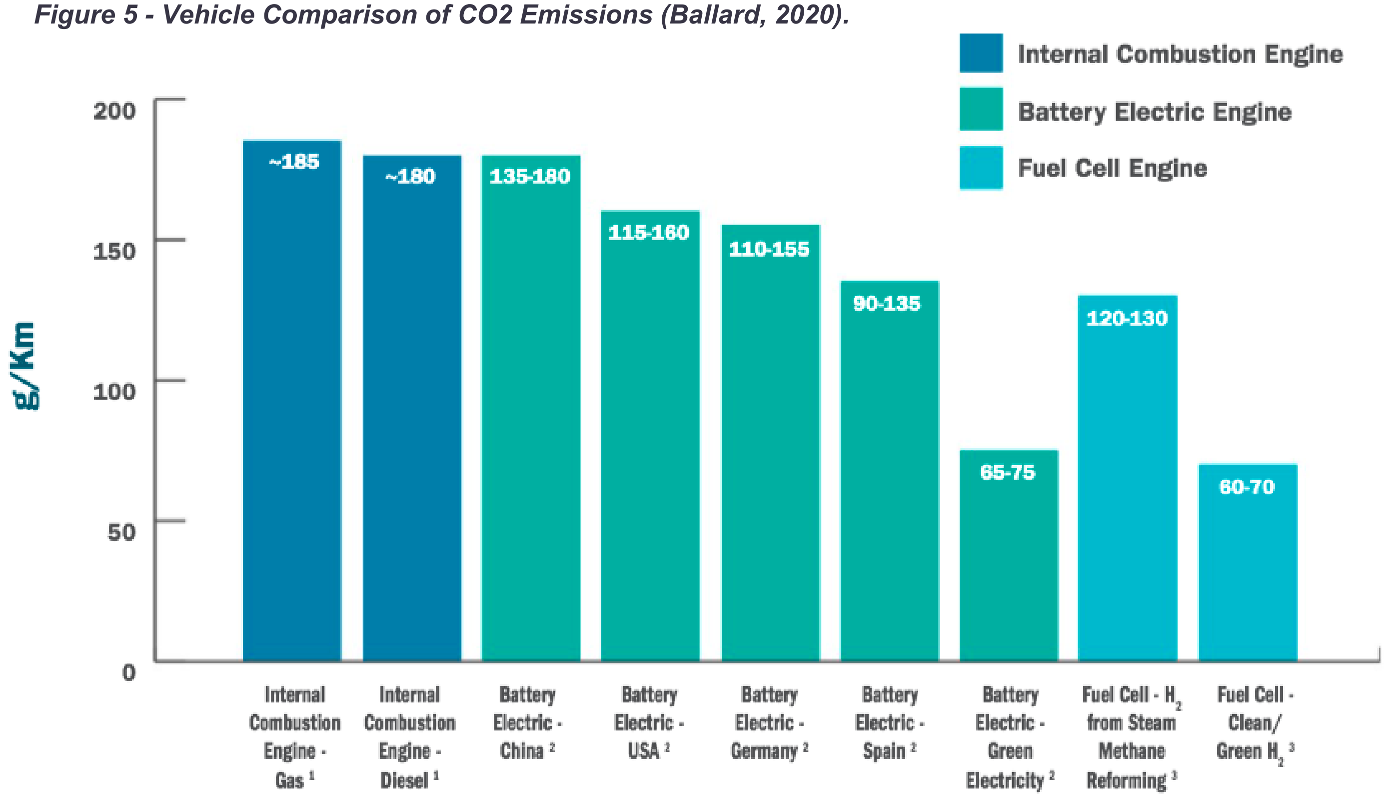

*Figure 4: Blue Ocean Strategy*

# Impact Analysis

Ballard Power Systems has a CSR score of 56 on CSR Hub meaning it is above the industry average. Ballard's 2019 ESG Report declares a commitment to becoming carbon neutral by 2030, dubbed "Mission Carbon Zero." The first step in reaching this goal for them was to evaluate and track their carbon emissions as well as perform Life-Cycle Assessments (LCA) of their products (Ballard, 2020). They hired independent consultant Offsetters Climate Solutions to perform an assessment of their company's emissions following the Greenhouse Gas Protocol Corporate Accounting and Reporting Standard, and they plan to report their emissions to the CDP.

The company itself exists in the alternative energy space with a purpose to decrease and eventually eliminate the environmental damage caused by transportation; their ESG Report touts hydrogen fuel cells as having less environmental impacts than internal combustion engines (ICE) or electric vehicles (EVs). The chart below shows a comparison of the life cycle emissions of ICE, EVs, and FCEVs from the manufacturing of the vehicles plus the emissions from driving them (Ballard, 2020).

***Figure 5 - Vehicle Comparison of CO2 Emissions (Ballard, 2020).***




Ballard's 2019 ESG Report outlines the U.N. Sustainable Development Goals that Ballard is focusing on to improve: affordable and clean energy; industry, innovation, and infrastructure; responsible production and consumption; climate action. The report highlights their strategy to minimize waste in their manufacturing, including refurbishing and recycling fuel cells at the end of their life. They also have deployed 42 initiatives to reduce their carbon footprint; these include promoting work-from-home & carpooling, collaborating with Toyota to get FCEVs for employees, switching to LEDs, recirculating hydrogen in testing, and installing regenerative load banks for electricity (Ballard, 2020).

While Ballard Canada decreased their Scope 2 & 3 emissions intensity from 2017 to 2018, their scope 1 emissions intensity increased by about 1000 $tCO_2$-equivalent. Emissions intensity is a way to calculate $CO_2e$ emissions per unit of activity, usually revenue. However, BLDP makes no mention of the unit they are using for their intensity calculations in their ESG Report suggesting that there could be greenwashing. Unfortunately, their absolute emissions are not included in the report making their claims suspect.

In the social responsibility section of their 2019 ESG Report, it seems that Ballard is behind in terms of female representation. Only 24% of their workforce is female and their board, senior leadership team, and mid-level management have even less representation. The executive level has the most female representation at only 33% (Ballard, 2020). Although below the mark of where companies ought to be on this issue, Ballard is ahead of the industry average and was named in the Globe & Mail "Women Lead Here" list which shows the top 73 Canadian companies leading this front. Another valuable social point in BLDP's ESG Report is that employees appreciated the company's environmental positioning in the employee survey. At least from an external point of view, their employees seem happy to work for a company with environmental values.

Part of the B Corp B Impact Assessment is to look at company diversity and involvement with the local community such as charitable donations (B Lab, 2019). Since BLDP has never made a net income, it makes sense that they do not donate to charity. However, they do host fundraisers to provide for charities such as United Way. Ballard also employs refugees on their Europe team. Another critical aspect of the B Impact Assessment is environmental impact of both the company's operations and the products. BLDP's products are designed for a market that aims to reduce transportation's emissions, so they score high for that measure. However, their operations emissions are only provided in terms of intensity as mentioned before, so they would likely not score very high in that measure.

Although BLDP is making headway in the zero-emission vehicle space and thus decreasing transportation emissions, upstream is another story. Based on results from the LCAs, one of Ballard's products- the fuel cell power module for the electric transit bus segment- is responsible for about 6 tons of $CO_2$ per unit. 68% of these life-cycle emissions are from the production of aluminum and platinum; however, 95% of the platinum is recycled (Ballard, 2020). Ballard's calculations must be based on blue or green hydrogen (hydrogen produced from fossil fuels & carbon-capture systems or from renewable energy, respectively). Currently, global production of hydrogen is mostly from fossil fuels without carbon-capture systems (grey hydrogen); not even 0.1% is produced from water electrolysis with renewable sources (IEA, 2019). Given hydrogen's coal and natural gas reliance, its production accounts for around 830 million metric tons of $CO_2$ annually (IEA, 2019). In order to support the adoption of hydrogen fuel cell vehicles that would benefit BLDP, these emissions would likely increase dramatically before the transition to renewables or implementation of carbon-capture systems are substantial. As mentioned earlier, the transport sector as a whole is responsible for about 8 billion metric tons of CO2e, so a substantial increase in grey hydrogen production emissions would almost nullify its climate benefits (Ritchie, et. al, 2020). If grey hydrogen production is not replaced by the production of blue or green hydrogen, then fuel cell buses will actually have 67% higher $CO_2$e emissions than diesel buses (Cox, et al., 2017).

The silver lining is that as renewable energy sources like wind and solar become less costly, green hydrogen can be produced by water electrolysis and enable a truly clean transportation sector. BLDP partnered with Deloitte China to reveal how hydrogen will be a key to our carbon neutral future; in the white paper they co-produced, they discuss the production of green hydrogen specifically and detail the costs involved (Cision, 2020). BLDP recognizes the need for green hydrogen over grey hydrogen and advocates its production.

# Financial Analysis and Success Indicators

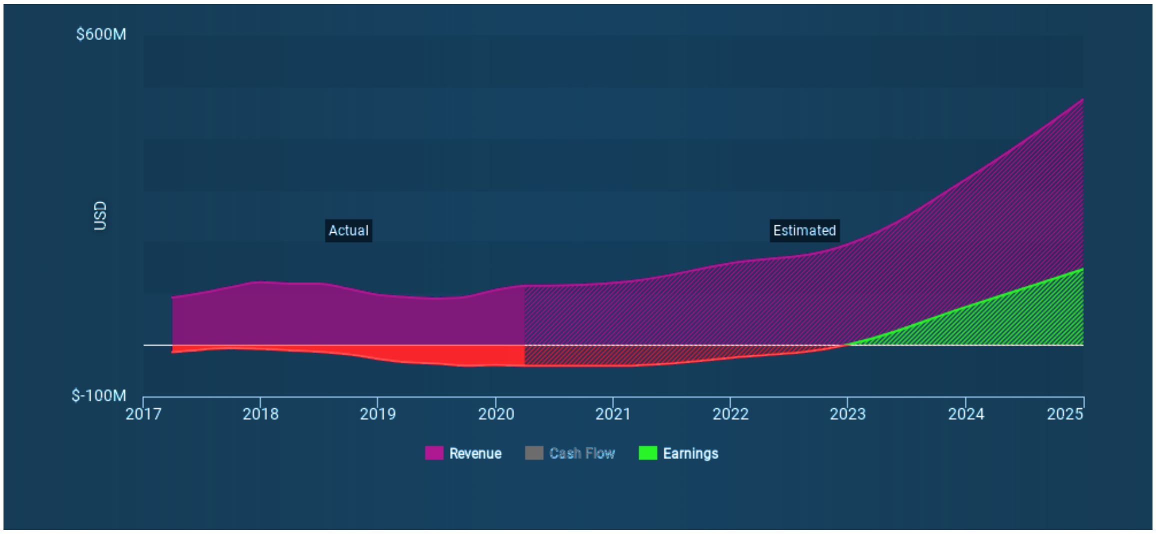
Ballard Power Systems has seen strong stock price growth in recent years. BLDP has been named to the *TSX30* (Toronto Stock Exchange top 30 performers) for 2 years in a row, recognizing their healthy 3-year share price growth (Cision, 2020). Their stock price has grown 166.4% over the past year and a whopping 808.5% in the past five years (Simply Wall St., 2020).

***Figure 6 Earnings & Revenue History – BLDP (Simply Wall St., 2020)***


Despite recent investor excitement, BLDP has yet to turn a profit in its 30+ years in the PEMFC industry. This can be viewed as a characteristic of the hydrogen fuel cell industry sector and doesn't imply that Ballard's management is doing a poor job; competitor Plug Power has also been in the red, and companies in this sector are eagerly anticipating the revolution of a "hydrogen economy." Although BLDP has been consistently in the red, their balance sheet has little debt burden as they have been relying heavily on equity investment. Their debt-to-assets ratio has regularly been well below the industry average of 50%, and currently BLDP is debt free (Simply Wall St., 2020). Plug Power on the other hand has a debt to equity ratio of 123% which is too high to be considered healthy

(Simply Wall St., 2020). Analysts predict that BLDP will finally turn a profit in 2022 with an annual growth rate of 54% (Simply Wall St., 2020). With high barriers of entry in the hydrogen fuel cell space, BLDP is well positioned with their intellectual property. Given that the hydrogen economy comes to life, BLDP will be able to capture a majority of the market share that will finally push them into growing profits.

*Figure 7 Future Growth – BLDP (Simply Wall St., 2020)*




BLDP saw their stock price skyrocket in the early 2000s as people anticipated the hydrogen economy. Unfortunately, the hydrogen bubble burst like the Hindenburg and people lost hope for hydrogen fuel cells. Now they are on the uptick again, and this time the world seems ready for it. Countries and companies the world over are committing to becoming carbon neutral, and many governments such as China are incorporating hydrogen fuel cells into their emissions reduction plan (Ballard, 2020).

*Figure 8 Historical Stock Price – BLDP (Simply Wall St., 2020)*



*Figure 9 Market Capitalization - BLDP (Macrotrends, 2020)*

Revenues for BLDP peaked in December of 2017, but they have been steadily rising to meet that mark again and are on track to surpass it (Simply Wall St., 2020). However, their earnings have been steadily decreasing. This may be due to their increased investments: BLDP's Property, Plant, and Equipment account has risen over 40% in the past two years and they have over $28 million in long-term investments (FactSet, 2020).

BLDP's market capitalization has seen a dramatic upswing over the past year (Macrotrends, 2020). This could be due to the embracing of hydrogen fuel cells as the future's key to zero emission heavy duty transportation. More than 60 countries have put hydrogen into their agenda towards decarbonizing. Also getting people's attention is the promising joint venture partnership with WeiChai in China (Cantech Letter, 2020).

# Analysis, Recommendations, and Conclusion

Ballard is on the right path of sustainability, but their social responsibility goals are lacking. They need to match their social goals with their environmental goals to remain relevant on the changing global stage. They need to increase the current 1:3 female to male representation in their workforce to about a 1:1 ratio. Ballard should also make stronger diversity and inclusion goals beyond gender ratios.

In order to frame their climate goals, Ballard should obtain Environmental Product Declarations (EPD) for all their products. For a company already pursuing life-cycle assessment data for their products, EPDs are the logical next step. EPDs have a third party verify the environmental claims and ensure transparency (The International EPD System). Another way to bolster their environmental claims would be to measure and publish their absolute emissions metrics rather than emissions intensity.

Ballard is currently a leader in hydrogen fuel cell technology. They must maintain this competitive edge if they want to continue capturing the momentum behind the hydrogen transportation industry. This is especially critical in China where the greatest potential for hydrogen is. China has been investing over $10 billion in its hydrogen transportation industry each year, and Ballard was wise to partner with WeiChai in a joint venture there. This partnership and their small office will enable them to grow their share of China's market, but they need to scale up their presence there if they want to maintain it. The conclusion is that Ballard should open its own manufacturing plant in China in the coming years.

One-way Ballard can help drive systems change is to contribute to the development of industry standards. The standards have been in flux, and more research is needed to nail them down and scale up the industry within those parameters. With thoroughly tested standards, the environmental impacts of the hydrogen industry can be better understood, and the benefits shared globally. Also, as a writer of standards, Ballard could gain a competitive advantage by advising them to match their planned product roadmap. As Ballard invents better fuel cell technology, they can guide the standards to become more stringent and ultimately grow demand for their product that meets the highest standards.

One major headwind the hydrogen transportation industry faces is the negative public perception of hydrogen as dangerous. However, hydrogen's safety is comparable to gasoline, and driving a FCEV is on par with an internal combustion engine vehicle (US DOE). To match public safety perception with reality, Ballard should promote the hydrogen transportation industry to end consumers. As a B2B company they normally do not advertise to end consumers, but the industry would grow faster and more steadily if the public were aware of its existence, environmental benefits, and safety.

Hydrogen-fueled transportation has come a long way since the Hindenburg, and Ballard is a forger of its future. Overall, Ballard is a relatively sustainable company with huge potential to make a dent in climate change. They are well positioned to capture a majority of the market share so long as they maintain their technological advantages, uphold their environmental claims, and raise the bar on social responsibility.

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